

# Pedro Ferreira

## Game Programmer

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## PROFESSIONAL SUMMARY

Game programmer with hands-on experience shipping titles across Unity, Unreal Engine, and custom C++/OpenGL engines. Strong foundation in engine architecture, real-time rendering, and performance optimization. Professional experience building gamification tools at Fujitsu with WebGL deployment. Comfortable across the full stack of game development, from low-level rendering pipelines and ECS frameworks to gameplay system.

## TECHNICAL SKILLS

**Languages:** C++, C#, Dart, Python, GLSL

**Engines & Frameworks:** Unity, Unreal Engine, Flame Engine, Custom C++/OpenGL Engine

**Graphics & Rendering:** OpenGL, Shader Programming, Real-time Rendering, Batch Rendering, WebGL

**Architecture:** ECS, Object Pooling, State Machines, Event-Driven Systems, Singleton Patterns

**Tools:** Git, Visual Studio, Performance Profiling, Android SDK

## WORK EXPERIENCE

### Game Programmer | Fujitsu Feb 2025 – Jul 2025

- Developed Unity-based 3D gamification tools used for employee training across Fujitsu teams.
- Implemented interactive tutorials, progress tracking systems, and achievement mechanics.
- Optimized WebGL builds to achieve stable 30 FPS in-browser on standard hardware.
- Collaborated with cross-functional teams to maintain clean, well-documented, production-ready code.

## PROJECTS

### Custom 2D Game Engine | C++, OpenGL, GLSL, SDL Sep 2024 – Feb 2025

[github.com/PedroFerreiraDevv/2d\\_game\\_engine](https://github.com/PedroFerreiraDevv/2d_game_engine)

- Engineered a 2D game engine from scratch with ECS architecture, hitting 60 FPS at 1080p.
- Implemented batch rendering pipeline, reducing draw calls by 80% compared to naive approach.
- Built core systems: sprite rendering, input handling, collision detection, and scene management.

### Xenon Clone – Arcade Shooter | C++, OpenGL, Custom Engine Sep 2024 – Feb 2025

- Built a complete vertical shooter on top of the custom engine with 3 enemy AI types and 4 weapon variants.
- Designed hierarchical state machines for enemy behavior and boss encounter patterns.
- Used object pooling to cut runtime memory allocations by 90%.

### Block Survival – Mobile Puzzle Game | Unity, C# 2025

[pedroferreiradev.itch.io/block-survival](https://pedroferreiradev.itch.io/block-survival)

- Developed a Tetris-style mobile puzzle game with custom touch input, bomb power-ups, and corruption mechanics.
- Built a custom DragTouchController using absolute position tracking for responsive mobile controls.
- Shipped WebGL build on itch.io; prepared for Google Play Store release.

### Freedom's Path – Stealth Game | Unity, C# Feb 2023 – Sep 2023

[itch.io \(500+ downloads\)](https://itch.io/500+downloads)

- Shipped a stealth game on itch.io with 500+ downloads and 4 different endings tied to collectibles.
- Implemented vision cone detection via raycasting, dynamic audio propagation, and state-based AI.

### Mobile 2D Shooter | Dart, Flame Engine Jan 2025 – Feb 2025

- Created a wave-based shooter with 20-wave progression and 5 power-up types.
- Optimized for 60 FPS on mid-range Android with object pooling handling 500+ concurrent entities.

### 3D Object Loader / Renderer | C++, OpenGL 2024

[GitHub](https://github.com)

- Built a 3D model loader and renderer using C++ and OpenGL with CS 1.6 assets.

### MIDI Parser for Rhythm Games | Python 2024

- Developed a MIDI file parser that outputs CSV formatted for easy integration with conventional game engines.

## EDUCATION

### Bachelor's Degree in Games and Multimedia 2021 – 2025

Polytechnic of Leiria, Portugal